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OPERATING SYSTEMS AND SYSTEMS PROGRAMMING (25CS2104E)

2026–27, Term-I

Project Problem Statement Submission Form

Section / Team Details

Section No.: 13

Team No.: 22

Project Title: Linux File Backup and Synchronization System Using System Calls

Team Members

Roll Number	Student Name	Signature
2500090225	B. Siddharatha	
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1. Abstract

Data loss due to accidental deletion, hardware failures, software crashes, and security threats is a common challenge in modern computing systems. Traditional manual backup methods are often inefficient, time-consuming, and prone to human error. To address this problem, this project proposes a

Linux-based File Backup and Synchronization System that automates the process of backing up files and maintaining consistency between source and destination directories.

The project applies key Operating Systems and System Programming concepts such as file management, process management, directory handling, file permissions, and inter-process communication. Linux system calls including open (), read (), write (), close (), stat (), mkdir(), opendir(), readdir(), fork (), and exec () will be used to access files, monitor changes, and perform backup operations efficiently.

The proposed solution will scan directories, identify newly created, modified, or deleted files, and synchronize them with a backup location while preserving file attributes and permissions. The system will also maintain activity logs and provide status reports to help users track backup operations.

The expected outcome is a reliable command-line utility that enables automatic file backup, synchronization, and restoration in a Linux environment. The system will improve data protection, reduce storage redundancy by copying only updated files, and demonstrate the practical application of Linux system calls and operating system concepts in developing efficient file management solutions.

2. Problem Statement

Modern computer systems generate and store large amounts of important data that are vulnerable to accidental deletion, hardware failures, software crashes, and unauthorized modifications. Many users rely on manual backup methods, which are often inconsistent, time-consuming, and prone to human error. Additionally, maintaining synchronization between original files and backup copies becomes challenging when files are frequently updated.

This project aims to develop a Linux-based File Backup and Synchronization System that automatically detects file changes and performs backup and synchronization operations efficiently. The system will ensure that updated files are copied to a backup location while preserving file integrity and minimizing redundant storage.

The project is highly relevant to Operating Systems and Systems Programming because it utilizes core OS concepts such as file management, process management, directory traversal, file permissions, and resource management. It also involves the use of Linux system calls and POSIX APIs for file access, directory handling, process creation, and system-level operations. Through this project, students gain practical experience in developing system software that interacts directly with the Linux operating system, demonstrating the real-world application of Operating Systems and Systems Programming concepts.

3. Objectives

- To develop a Linux-based system that automatically performs file backup and synchronization between source and destination directories.
- To detect file creation, modification, and deletion events and update backup copies efficiently using Linux system calls.
- To implement secure file management by preserving file permissions, attributes, and directory structures during backup operations.
- To demonstrate the practical application of Operating Systems and Systems Programming concepts such as file handling, process management, directory traversal, and Linux system calls.

4. Proposed Methodology

The project will be developed as a Linux command-line application using C/C++ and Linux system calls. The system will allow users to specify a source directory and a backup destination directory. Initially, the application will scan the source directory and create a complete backup of all files and subdirectories.

The system will then periodically monitor the source directory for changes such as file creation, modification, and deletion. Using Linux file and directory handling system calls such as open (), read (), write (), stat (), opendir (), and readdir (), the application will identify differences between the source

and backup locations. Only new or modified files will be copied to reduce storage usage and improve performance.

The implementation may use separate processes through fork () and exec () to perform backup tasks and generate activity logs simultaneously. The system will preserve file permissions, timestamps, and directory structures during synchronization. All backup operations, errors, and synchronization activities will be recorded in log files for monitoring and recovery purposes.

Flow of the System:

1. User specifies source and backup directories.
2. System scans the source directory and creates an initial backup.
3. Files and directories are compared using metadata and timestamps.
4. New or modified files are synchronized to the backup location.
5. Deleted files are handled according to synchronization rules.
6. Backup activities are logged, and status reports are generated.

The final system will provide an efficient, reliable, and automated solution for file backup and synchronization in a Linux environment.

5. Operating Systems Concepts / Linux APIs Used

OS Concept / Linux API / System Call	Purpose in the Project
File Management	Manage creation, access, modification, and backup of files.
Directory Management	Traverse source and backup directories and maintain directory structure.
Process Management	Create and manage processes for backup and synchronization tasks.
File permissions	Preserve file access permissions during backup and restoration.
fork ()	Create child processes for concurrent backup tasks.
exec ()	Execute synchronization or utility programs when needed.
POSIX File APIs	Provide standardized file and directory handling across Linux systems.
Logging Mechanism	Record backup activities, errors, and synchronization status.

6. Individual Contribution

Roll Number	Student Name	Individual Responsibility

2500090225	B. Siddharatha	Design and develop the core backup system using Linux file-related system calls.
2520090027	G. Bhupathi	Implement directory synchronization, change detection, and file comparison mechanisms.
2520080006	G V Karthik Reddy	Perform testing, logging, debugging, GitHub management, and project
		documentation.

7. Tools / Platforms / Software Used

Tool / Platform / Software	Purpose
Linux / Ubuntu	Operating system used for development, testing, and execution of the project.
C / C++	Programming language used to implement the backup and synchronization system.
GCC (GNU Compiler Collection)	Compile and build the source code.
POSIX APIs	Provide standardized system-level functions for file and process management.
Linux System Calls	Enable file handling, directory traversal, process creation, and synchronization operations.
Terminal / Bash Shell	Execute commands, run programs, and test functionality.
Make / Makefile	Automate compilation and build processes.
Log Files	Store backup activities, synchronization status, and error reports.
GitHub	Repository hosting, collaboration, and project tracking.

8. Expected Outcome

The developed system will automatically back up and synchronize files between a source directory and a backup directory in a Linux environment. It will detect file additions, modifications, and deletions, perform the necessary synchronization operations, preserve file permissions and directory structures, and maintain logs of all backup activities for monitoring and recovery purposes.

The outcome will be a reliable command-line utility that improves data protection, reduces the risk of data loss, and minimizes manual effort in managing backups. The project will also demonstrate the practical application of Linux system calls, POSIX APIs, file management, and process management concepts in solving real-world file synchronization and backup problems.

9. GitHub Repository

GitHub Repository Link: https://github.com/Bhupathi22/Linux-File-Backup-and-Synchronization-System-Using-System-Calls-OSSP-Project-

Faculty Name and Signature

Faculty Name: Dr. Alekhya

Signature: _____

Remarks: _____

Faculty Use

Project Approved: _____

☐ Yes ☐ No

Date:

Faculty Signature: _____